# Adaptive-query lower bounds for RE-03

**Abstract**

For the HODLR approximation problem RE-03, this note develops a lower bound

$$q_*(n, k, \varepsilon) \geq c \min\{n,\ k(L+1) + k/\varepsilon^2\}, \qquad n = 2^L k,$$

with a universal constant $c > 0$. The argument applies to arbitrary measurable, randomized adaptive matrix–vector query algorithms, with exact real arithmetic and access to both $A$ and $A^\mathsf{T}$. It combines an exact-recovery obstruction with a finite packing of rank-$k$ projectors, a shifted-GOE transcript bound, and a deterministic reduction from approximation excess to subspace error. Together with the $n$-query exact-projection algorithm, it gives $q_* = \Theta(n)$ when $\varepsilon \leq \sqrt{k/n}$. The argument and its constants are provided in full, with numerical sanity checks in the accompanying code.

**Scope and status.** This is an unreviewed partial research result, not a complete solution of RE-03. In particular, it does not give matching upper and lower bounds for all $(k, L, \varepsilon)$. The new lower-bound proof below has not been independently or formally verified. The numerical checks test selected identities and constructions; they do not certify the adaptive lower bound.

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## 1 Problem and principal statements

Let $n = 2^L k$, where $k \geq 1$, $L \geq 2$ are integers, and let $0 < \varepsilon < 1/2$. The class $\mathcal{H}_{n,k}$ consists of real square matrices whose two off-diagonal sibling blocks at each node of the prescribed balanced binary partition have rank at most $k$. The diagonal children satisfy the same condition recursively; the terminal $k \times k$ blocks are unrestricted. Write

$$\mathrm{OPT}(A) = \min_{H \in \mathcal{H}_{n,k}} \|A - H\|_{\mathsf{F}}.$$

The query model permits one vector product $Av$ or $A^{\mathsf{T}}v$ per call, with the vector and side chosen adaptively. Only calls are charged. The algorithm must output a member $B \in \mathcal{H}_{n,k}$ satisfying

$$\mathbb{P}\{\|A - B\|_{\mathsf{F}} \leq (1+\varepsilon)\,\mathrm{OPT}(A)\} \geq 0.99 \quad \text{for every fixed } A. \tag{1}$$

This is the model and quantifier order in the RE-03 specification.[1]

The published comparison point is the upper bound $O(\min\{n, kL^4/\varepsilon^3\})$. The cited lower bound is $\Omega(kL + k/\varepsilon)$ in the specified regime $n \geq c_0 k/\varepsilon$, together with the unrestricted exact-recovery obstruction $\Omega(kL)$.[2]

**Theorem 1.1** (Principal lower bound)**.** *There is a universal constant $c > 0$ such that, for every admissible $(n, k, \varepsilon)$,*

$$q_*(n,k,\varepsilon) \geq c \min\left\{n,\ k(L+1) + \frac{k}{\varepsilon^2}\right\}. \tag{2}$$

*More specifically,*

$$q_*(n,k,\varepsilon) \geq k(L+1), \qquad q_*(n,k,\varepsilon) \geq c_* \min\left\{n, \frac{k}{\varepsilon^2}\right\}, \tag{3}$$

*where one conservative choice is*

$$D = 2^{24}, \qquad c_* = (6400D^2)^{-1}, \qquad c = c_*/2.$$

The explicit constants are deliberately loose. In particular, this theorem does not assert a large numerical fraction of $n$ queries for moderate-sized instances. Its claim is a uniform asymptotic rate.

**Corollary 1.2** (A fully characterized regime)**.** *For $0 < \varepsilon \leq \sqrt{k/n}$,*

$$c_* n \leq q_*(n,k,\varepsilon) \leq n. \tag{4}$$

*Thus $q_*(n,k,\varepsilon) = \Theta(n)$ in this regime, with universal constants.*

The hard inputs used for the accuracy lower bound are symmetric, although the permitted approximations need not be symmetric. Symmetry is useful because $Av$ and $A^{\mathsf{T}}v$ coincide on these inputs. The proof therefore does not lose validity when the algorithm is allowed to choose either side.

[1] RE-03, *Optimal matvec query complexity of HODLR approximation*, problem statement and audit of September 10, 2026; see source [1].

[2] Chen, Duman Keles, Halikias, Musco, Musco, and Persson, *Near-optimal hierarchical matrix approximation from matrix-vector products*, Theorems 1.1–1.2 and Section 6, arXiv v2; see source [2].

## 2 Block geometry and exact recovery

### 2.1 The objective separates over the partition

Let $\mathcal{R}$ denote all *ordered* off-diagonal sibling blocks $(I, J)$ in the binary partition. Thus both $(I, J)$ and $(J, I)$ occur. Let $\mathcal{D}$ denote the terminal diagonal blocks. These blocks partition all entries of an $n \times n$ matrix. For a rectangular matrix $C$, define

$$\tau_k(C)^2 = \sum_{i>k} \sigma_i(C)^2.$$

The singular values are listed in nonincreasing order. An empty tail has value zero.

**Lemma 2.1** (Separable optimum and nonnegative local excess)**.** *For every real $A$,*

$$\mathrm{OPT}(A)^2 = \sum_{(I,J)\in\mathcal{R}} \tau_k(A_{I,J})^2. \tag{5}$$

*Moreover, if $B \in \mathcal{H}_{n,k}$, then*

$$\begin{aligned}\|A - B\|_{\mathsf{F}}^2 - \mathrm{OPT}(A)^2 = &\sum_{(I,J)\in\mathcal{R}} \left(\|A_{I,J} - B_{I,J}\|_{\mathsf{F}}^2 - \tau_k(A_{I,J})^2\right) \\ &+ \sum_{I\in\mathcal{D}} \|A_{I,I} - B_{I,I}\|_{\mathsf{F}}^2\,,\end{aligned} \tag{6}$$

*and every summand on the right is nonnegative.*

*Proof.* Squared Frobenius norms add over disjoint sets of entries. The rank constraint on each off-diagonal block is independent of every other such constraint. For a block $C$, its best rank-$k$ approximation is its truncated SVD, with squared error $\tau_k(C)^2$. One way to see this is to choose a projector $Q$ onto a subspace containing the row space of any proposed rank-$k$ matrix $D$. Then

$$\|C - D\|_{\mathsf{F}}^2 \geq \|C(I - Q)\|_{\mathsf{F}}^2 = \|C\|_{\mathsf{F}}^2 - \mathrm{tr}(QC^{\mathsf{T}}C).$$

The last trace is at most the sum of the $k$ largest eigenvalues of $C^{\mathsf{T}}C$. Equality is attained by its leading right singular subspace. The unrestricted leaf blocks can be copied exactly. This proves (5), and subtraction gives (6). ☐

**Proposition 2.2** (The $n$-query baseline)**.** *For every admissible triple, $q_*(n, k, \varepsilon) \leq n$.*

*Proof.* Query $Ae_i$ for $i = 1, \ldots, n$, thereby recovering every column of $A$. Copy the leaf blocks and take an exact truncated SVD on every off-diagonal sibling block. Lemma 2.1 shows that the resulting $B$ attains $\mathrm{OPT}(A)$ exactly. No additional oracle calls are needed. ☐

### 2.2 An adaptive-safe dimension obstruction

A dimension count must be justified carefully in an exact-real model with arbitrary measurable computation. Counting the number of real coordinates in a transcript alone is not sufficient. Here a Gaussian conditional-law argument supplies the needed justification.

**Proposition 2.3** (Exact-recovery lower bound)**.** *Every algorithm satisfying* (1) *has worst-case query count at least $k(L + 1)$.*

*Proof.* Consider the linear subspace $\mathcal{V} \subset \mathcal{H}_{n,k}$ in which each ordered off-diagonal block is zero outside its first $k$ columns, while those columns and all terminal diagonal blocks are unrestricted. At each of the $L$ levels, the free off-diagonal entries number $nk$. The leaves contribute another $nk$. These supports are disjoint, so

$$d := \dim \mathcal{V} = nk(L+1). \tag{7}$$

Put independent standard normal coordinates on a fixed basis of this subspace. Condition first on the algorithm's random seed. Given the preceding transcript, the next vector and side are fixed, so its answer is a linear map of the unknown $d$-dimensional Gaussian coordinate vector, with rank at most $n$.

Inductively, after $q$ calls the conditional distribution is a Gaussian on an affine subspace of dimension at least $d - nq$. This follows by conditioning a Gaussian on linear equations at each stage; the measurement map at that stage is fixed once the prior answers are conditioned upon. In particular, if $nq < d$, the posterior is non-atomic. Any proposed output, being a function of the transcript and seed, has conditional probability zero of equaling the input. Integrating over transcripts and seeds leaves probability zero of exact recovery.

But every input from $\mathcal{V}$ has $\mathrm{OPT}(A) = 0$, and hence (1) requires exact recovery with probability at least 0.99 for every such input. Its average success under the Gaussian input distribution would then be at least 0.99, a contradiction. Thus $q \geq d/n = k(L+1)$. □

## 3 Information in adaptive shifted-GOE queries

For a symmetric matrix $G$, use the GOE normalization

$$G_{ij} = G_{ji} \sim N(0, 1/n) \quad (i < j), \qquad G_{ii} \sim N(0, 2/n), \tag{8}$$

with independent upper-triangular entries. Its density on the vector space of symmetric matrices is proportional to $\exp(-n \|G\|_{\mathsf{F}}^2 /4)$. In particular, it is isotropic for the Frobenius inner product on that vector space.

**Lemma 3.1** (Adaptive transcript divergence)**.** *Fix symmetric $A_0, S \in \mathbb{R}^{n\times n}$ with $\|S\|_{\mathrm{op}} \leq 1$ and a real $\delta$. Let $\mathbb{P}_S$ be the law of the transcript of any randomized adaptive algorithm using at most $q \leq n$ vector queries on*

$$A = A_0 + \delta S + G.$$

*Let $\mathbb{P}_0$ be its transcript law on $A_0 + G$. Include the independent random seed in the transcript. Then*

$$D_{\mathrm{KL}}(\mathbb{P}_S \| \mathbb{P}_0) \leq \frac{n\delta^2 q}{2}. \tag{9}$$

*The statement permits queries on either side of $A$.*

*Proof. Reduction to orthonormal fresh directions.* All matrices in question are symmetric, so choosing the transpose side gives no additional type of measurement. Subtract the known product $A_0 v$ from each answer. If a new query vector has a component in the span of previous queries, its image on that component is already known. Removing it and normalizing the remaining vector recovers the same fresh information using one call. Queries wholly in the old span give no information and can be omitted. We may therefore work with an orthonormal sequence of fresh vectors, chosen adaptively. An algorithm that stops early can be padded with orthogonal queries. Data processing then bounds its original transcript by the padded transcript. Fix the seed for the moment.

*Gaussian exposure under adaptivity.* After $t-1$ fresh directions, let $V$ be the matrix of those directions and $\Pi = VV^\top$. For a fixed value of $S$, the observed answers determine $Z = GV$. The symmetric component of $G$ determined by these observations is

$$G_{\mathrm{obs}} = ZV^\top + VZ^\top - V(V^\top Z)V^\top. \tag{10}$$

The undetermined part is $(I-\Pi)G(I-\Pi)$. These are orthogonal subspaces of the space of symmetric matrices for the Frobenius inner product. Conditional on the history, the latter part remains a centered GOE compression on range$(I-\Pi)$, with the normalization $1/n$, not the inverse of the remaining dimension.

For completeness, this claim also holds when $V$ is adaptive. Initially it is just the orthogonal decomposition of an isotropic Gaussian. Given the past, the next direction is fixed. In an orthonormal basis of the remaining space beginning with that direction, the new Gaussian column and the unexposed lower-right symmetric block are independent. Conditioning on the column leaves that block unchanged. Induction proves the assertion. The choice of the next vector is a function of already conditioned-on answers, so it does not impose an additional constraint on the unexplored block.

*One-step divergence.* Let $v$ be the next unit vector, with $\Pi v = 0$. The new information is the component

$$y = (I-\Pi)(A-A_0)v.$$

The component in range$(\Pi)$ is already determined by the old answers and symmetry: $V^\top(A-A_0)v = ((A-A_0)V)^\top v$. At a fixed history, the conditional mean of $y$ under the shifted law is $\delta(I-\Pi)Sv$, and under the baseline it is zero. In both cases its conditional covariance, restricted to range$(I-\Pi)$, is

$$\Sigma_v = \frac{1}{n}((I-\Pi) + vv^\top). \tag{11}$$

The variance is $2/n$ in direction $v$ and $1/n$ in the orthogonal remaining directions. In particular, $\Sigma_v^{-1} \preceq nI$ on its support.

Two Gaussians with identical covariance and mean difference $m$ have divergence $m^\top\Sigma_v^{-1}m/2$, as follows directly by integrating the logarithm of their density ratio. The conditional divergence is therefore at most

$$\frac{n\delta^2}{2}\,\|(I-\Pi)Sv\|^2 \le \frac{n\delta^2}{2}.$$

Applying the chain rule for relative entropy to the successive fresh answers and summing gives (9). Integrating over the seed gives the same bound for randomized algorithms, since its distribution is identical in both experiments. □

*Remark* 3.2 (Why exact arithmetic does not invalidate the bound)*.* This is a divergence calculation for the laws of exact real-valued answers. No rounding, finite-precision hypothesis, bounded query coefficients, or nonadaptive restriction is used. Normalizing a query loses no information because its scale is known. The unexplored Gaussian compression is what prevents arbitrary measurable postprocessing from revealing the hidden parameter for free.

## 4 A local approximation-to-subspace inequality

The next lemma is the deterministic part of the construction. It handles arbitrary perturbations, nonsymmetric estimates, and estimates of rank strictly smaller than $k$.

**Lemma 4.1** (Local loss)**.** *Let $P$ be a rank-$k$ orthogonal projector on $\mathbb{R}^{2k}$, and let $0 < \delta \le 1/2$. Put*

$$C = (1-\delta)I + 2\delta P.$$

*For any $W \in \mathbb{R}^{2k\times 2k}$ and any matrix $D$ of rank at most $k$, choose a rank-$k$ orthogonal projector $Q$ whose range contains the row space of $D$. Then*

$$\begin{aligned} &\|C+W-D\|_{\mathsf{F}}^2 - \tau_k(C+W)^2 \\ &\quad \ge \delta\,\|P-Q\|_{\mathsf{F}}^2 - \left(\frac{(1+\delta)^2}{\delta} + 1\right)\|W\|_{\mathsf{F}}^2 . \end{aligned} \tag{12}$$

*Proof.* Write $d = \|P-Q\|_{\mathsf{F}}$ and $w = \|W\|_{\mathsf{F}}$. Since $D = DQ$,

$$\|C+W-D\|_{\mathsf{F}}^2 \ge \|(C+W)(I-Q)\|_{\mathsf{F}}^2 .$$

The rank-$k$ matrix $(1+\delta)P = CP$ is a valid competitor, so

$$\tau_k(C+W)^2 \le \|C(I-P)+W\|_{\mathsf{F}}^2 .$$

Subtracting these inequalities and expanding squared norms yields

$$\|C(I-Q)\|_{\mathsf{F}}^2 - \|C(I-P)\|_{\mathsf{F}}^2 + 2\,\langle W, C(P-Q)\rangle_{\mathsf{F}} - \|WQ\|_{\mathsf{F}}^2 . \tag{13}$$

Because $C^2 = (1-\delta)^2 I + 4\delta P$, the clean difference is

$$\operatorname{tr}((P-Q)C^2) = 4\delta(k - \operatorname{tr}(PQ)) = 2\delta d^2.$$

The cross term is bounded below by $-2(1+\delta)wd$, and the last term is at least $-w^2$. Thus (13) is bounded below by

$$2\delta d^2 - 2(1+\delta)wd - w^2.$$

Use $2(1+\delta)wd \le \delta d^2 + (1+\delta)^2 w^2/\delta$ to obtain (12). □

The projector $Q$ can be selected measurably from $D$: choose a deterministic orthonormal basis of its row space, then extend it by a fixed coordinate-based Gram–Schmidt rule until its dimension is $k$. No optimality, symmetry, or uniqueness assumption on $D$ is needed.

# 5 An explicit finite packing

All logarithms in this section and in the information calculation are natural. The metric on tuples of projectors is the square root of the sum of their squared Frobenius distances.

**Lemma 5.1** (Projector tuples)**.** *For every pair of positive integers $k, M$, there is a finite collection*

$$\{(P_1^\theta, \dots, P_M^\theta) : \theta = 1, \dots, N\}$$

*of rank-$k$ orthogonal projectors on $\mathbb{R}^{2k}$ such that*

$$\log N \ge \frac{Mk^2}{200} \tag{14}$$

*and, whenever $\theta \ne \theta'$,*

$$\sum_{j=1}^{M} \left\| P_j^\theta - P_j^{\theta'} \right\|_{\mathsf{F}}^2 \ge \frac{Mk}{1600}. \tag{15}$$

*Proof. Step 1: a single-block alphabet.* Suppose first that $k \geq 4$. Let $S$ range over sign matrices with independent entries $\pm 1/\sqrt{k}$. For fixed unit vectors $x, y$, the random variable $x^{\mathsf{T}} S y$ has moment-generating function at most $\exp(t^2/(2k))$, because $\cosh u \leq \exp(u^2/2)$ and $\sum_{ij} x_i^2 y_j^2 = 1$. Thus

$$\mathbb{P}\{|x^{\mathsf{T}} S y| > a\} \leq 2\exp(-ka^2/2).$$

A 1/4-net of the unit sphere has size at most $9^k$: take a maximal separated set and compare disjoint balls of radius 1/8 with the enclosing ball of radius 9/8. For nets in both variables, $\|S\|_{\mathrm{op}} \leq 2 \max_{x,y \text{ in nets}} |x^{\mathsf{T}} S y|$. Consequently,

$$\mathbb{P}\{\|S\|_{\mathrm{op}} > 10\} \leq 2\exp(-(25/2 - 2\log 9)k) < 1/2.$$

At least $2^{k^2-1}$ sign matrices are therefore good.

Let $h(p) = -p\log p - (1-p)\log(1-p)$. A binary Hamming ball of radius $\lfloor k^2/4 \rfloor$ has size at most $\exp(h(1/4)k^2)$. To verify the usual binomial bound, multiply each summand by $p^i(1-p)^{d-i}$ with $d = k^2$ and $p = 1/4$; for $i \leq pd$ this weight is at least $\exp(-dh(p))$, while the weighted sum is at most one. Greedy deletion of such balls from the good matrices gives a set with mutual Hamming distance greater than $k^2/4$ and log cardinality at least

$$(\log 2 - h(1/4))k^2 - \log 2 \geq k^2/20, \qquad k \geq 4.$$

For each retained sign matrix set $T = S/20$, so $\|T\|_{\mathrm{op}} \leq 1/2$, and let $P_T$ project onto the graph of $T$, the column space of $[I\ T^{\mathsf{T}}]^{\mathsf{T}}$. For two matrices $T, U$, an orthonormal basis of the graph of $U$ is $[I\ U^{\mathsf{T}}]^{\mathsf{T}}(I + U^{\mathsf{T}}U)^{-1/2}$; an orthonormal basis of the orthogonal complement of the graph of $T$ is $[-T\ I]^{\mathsf{T}}(I + TT^{\mathsf{T}})^{-1/2}$. Hence

$$\frac{1}{2}\|P_T - P_U\|_{\mathsf{F}}^2 = \left\|(I + TT^{\mathsf{T}})^{-1/2}(U - T)(I + U^{\mathsf{T}}U)^{-1/2}\right\|_{\mathsf{F}}^2. \tag{16}$$

Both inverse square roots have minimum singular value at least $1/\sqrt{1 + 1/4}$. The Hamming separation also gives $\|T - U\|_{\mathsf{F}}^2 > k/400$. It follows that

$$\|P_T - P_U\|_{\mathsf{F}}^2 \geq \frac{2}{(1 + 1/4)^2}\|T - U\|_{\mathsf{F}}^2 \geq k/400.$$

For $k = 1, 2, 3$, instead take the two complementary coordinate projectors. Their squared distance is $2k$ and their log cardinality is $\log 2 \geq k^2/20$. Thus for every $k$ there is a single-block alphabet of size $Q \geq 2$ with $\log Q \geq k^2/20$ and squared separation at least $k/400$.

*Step 2: a product code.* In the $Q$-ary cube of length $M$, a Hamming ball of radius $\lfloor M/4 \rfloor$ has size at most

$$\exp\big(Mh(1/4) + (M/4)\log(Q-1)\big).$$

Indeed, factor $(Q-1)^{M/4}$ out of the relevant binomial sum and apply the binary bound just proved. Greedy packing then gives mutual Hamming distance greater than $M/4$ and

$$\log N \geq M\big(\log Q - h(1/4) - \tfrac{1}{4}\log(Q-1)\big) \geq \tfrac{1}{10}M\log Q \geq Mk^2/200.$$

The middle inequality holds for $Q \geq 2$: the function $0.9\log Q - 0.25\log(Q-1)$ is increasing there, and its value at 2 is greater than $h(1/4)$. Each differing coordinate contributes at least $k/400$ to squared projector distance. This proves (15). $\square$

## 6 The hard distribution and the accuracy lower bound

### 6.1 Matrices indexed by the packing

Set $M = n/(4k) = 2^{L-2}$. Partition the coordinates into $M$ consecutive nodes of size $4k$, and split each into two sets $I_j, J_j$ of size $2k$. These are actual sibling blocks of the HODLR partition. Crucially, they are not the unrestricted $k \times k$ leaf blocks.

Take the packing from Lemma 5.1, and put

$$\begin{aligned} R_j^\theta &= 2P_j^\theta - I_{2k}, \\ A_0 &= \operatorname{blockdiag}_{j=1}^M \begin{bmatrix} 0 & I_{2k} \\ I_{2k} & 0 \end{bmatrix}, \\ S_\theta &= \operatorname{blockdiag}_{j=1}^M \begin{bmatrix} 0 & R_j^\theta \\ R_j^\theta & 0 \end{bmatrix}, \qquad A = A_0 + \delta S_\theta + G. \end{aligned} \tag{17}$$

Each $R_j^\theta$ is a symmetric orthogonal matrix. Thus $S_\theta$ is symmetric orthogonal too, and $\|S_\theta\|_{\mathrm{op}} = 1$. The upper targeted block of the clean matrix is

$$C_j = (1-\delta)I_{2k} + 2\delta P_j^\theta.$$

Its singular values are $1+\delta$ and $1-\delta$, each repeated $k$ times. The lower targeted block is its transpose. All other clean blocks are zero. By Lemma 2.1,

$$\mathrm{OPT}(A_0 + \delta S_\theta)^2 = 2Mk(1-\delta)^2 = \frac{n}{2}(1-\delta)^2. \tag{18}$$

The packing parameters become

$$\log N \geq \frac{nk}{800}, \qquad \sum_j \left\| P_j^\theta - P_j^{\theta'} \right\|_{\mathsf{F}}^2 \geq \frac{n}{6400}. \tag{19}$$

### 6.2 A good-noise event

Define

$$Z = \|G\|_{\mathsf{F}}^2, \qquad T = \sum_{j=1}^M \left\| G_{I_j, J_j} \right\|_{\mathsf{F}}^2.$$

The GOE normalization gives

$$\mathbb{E}Z = n + 1 \leq \tfrac{5}{4}n, \qquad \mathbb{E}T = k.$$

For the second identity, there are $M(2k)^2 = nk$ entries in the targeted *upper* blocks, each with variance $1/n$. These entries are disjoint. Markov's inequality and a union bound imply that the event

$$\mathcal{E} = \{Z \leq 10n,\ T \leq 10k\} \tag{20}$$

has probability at least

$$\mathbb{P}(\mathcal{E}) \geq 1 - \frac{1}{8} - \frac{1}{10} = \frac{31}{40}. \tag{21}$$

On this event, use the best clean HODLR approximation as a competitor for the noisy matrix and apply $\|X+Y\|_{\mathsf{F}}^2 \leq 2\|X\|_{\mathsf{F}}^2 + 2\|Y\|_{\mathsf{F}}^2$:

$$\mathrm{OPT}(A)^2 \leq n(1-\delta)^2 + 2Z \leq 21n. \tag{22}$$

No independence between $Z$ and $T$ is needed.

### 6.3 Decoding the hidden projectors

Suppose an approximation algorithm succeeds on an input from (17). Choose $Q_j$ to be a rank-$k$ projector containing the row space of $B_{I_j,J_j}$, using the measurable rule following Lemma 4.1. Let

$$r^2 = \sum_{j=1}^{M} \left\| P_j^\theta - Q_j \right\|_{\mathsf{F}}^2 .$$

Since $\varepsilon < 1/2$, success and (22) imply, on $\mathcal{E}$,

$$\begin{aligned} \|A - B\|_{\mathsf{F}}^2 - \mathrm{OPT}(A)^2 &\le (2\varepsilon + \varepsilon^2)\,\mathrm{OPT}(A)^2 \\ &\le \frac{105}{2}\varepsilon n. \end{aligned} \tag{23}$$

Sum Lemma 4.1 over the targeted upper blocks. Their excesses are nonnegative and sum to at most the global excess by Lemma 2.1. Also,

$$\frac{(1+\delta)^2}{\delta} + 1 = \frac{1 + 3\delta + \delta^2}{\delta} \le \frac{3}{\delta}, \qquad 0 < \delta \le 1/2.$$

It follows that

$$r^2 \le \frac{105\varepsilon n}{2\delta} + \frac{30k}{\delta^2}. \tag{24}$$

Now set

$$D = 2^{24}, \qquad \varepsilon_0 = \frac{1}{2D}, \qquad \delta = D\varepsilon, \tag{25}$$

and first restrict to

$$0 < \varepsilon \le \varepsilon_0, \qquad n \ge k/\varepsilon^2. \tag{26}$$

Then $0 < \delta \le 1/2$ and

$$\frac{r^2}{n} \le \frac{105}{2D} + \frac{30}{D^2} < \frac{1}{102400} < \frac{1}{25600}. \tag{27}$$

The smallest squared separation in (19) is at least $n/6400$. Thus $r$ is less than half the minimum separation. Nearest-neighbor decoding of the tuple $(Q_1, \ldots, Q_M)$ in the finite packing recovers $\theta$ uniquely. This is a measurable operation, and no query is needed to do it.

Let $\theta$ be uniform on the packing, independently of $G$ and the algorithm's seed. The fixed-input guarantee (1) implies average success at least 0.99 under this distribution. The success event and $\mathcal{E}$ therefore intersect with probability at least

$$\frac{31}{40} - \frac{1}{100} = \frac{153}{200}. \tag{28}$$

Hence the decoding error probability is at most $47/200$.

### 6.4 Information cannot support that decoding probability

Let $\mathcal{T}$ denote the transcript including the seed. By Lemma 3.1, and using the baseline transcript law generated by $A_0 + G$,

$$I(\theta; \mathcal{T}) \le \frac{1}{N} \sum_{\theta=1}^{N} D_{\mathrm{KL}}(\mathbb{P}_\theta \| \mathbb{P}_0) \le \frac{n\delta^2 q}{2}. \tag{29}$$

The first inequality follows from the identity

$$\frac{1}{N}\sum_{\theta} D_{\mathrm{KL}}(\mathbb{P}_\theta \| \mathbb{P}_0) = I(\theta; \mathcal{T}) + D_{\mathrm{KL}}(\overline{\mathbb{P}} \| \mathbb{P}_0),$$

where $\overline{\mathbb{P}}$ is the mixture transcript law.

We recall the needed entropy bound directly. If $\widehat{\theta}$ is any transcript-based decoder and $p_e = \mathbb{P}(\widehat{\theta} \neq \theta)$, conditioning also on the error indicator gives

$$H(\theta \mid \mathcal{T}) \leq \log 2 + p_e \log N.$$

Since $H(\theta) = \log N$, this yields Fano's inequality in the form

$$p_e \geq 1 - \frac{I(\theta; \mathcal{T}) + \log 2}{\log N}. \tag{30}$$

Assume, for a contradiction, that

$$q \leq \frac{k}{6400\delta^2}. \tag{31}$$

This bound is less than $n$ under (26), so the hypothesis $q \leq n$ in Lemma 3.1 applies. Equations (29) and (19) give $I(\theta; \mathcal{T}) \leq (\log N)/16$. Moreover, (26) implies $nk \geq 4D^2$, so $\log N \geq nk/800 \geq 16 \log 2$. Equation (30) therefore gives $p_e \geq 7/8$, contradicting $p_e \leq 47/200$ from (28). We have proved the following.

**Proposition 6.1** (Accuracy lower bound in the core regime)**.** *If $\varepsilon \leq \varepsilon_0$ and $n \geq k/\varepsilon^2$, then*

$$q_*(n, k, \varepsilon) \geq c_* \frac{k}{\varepsilon^2}, \qquad c_* = (6400D^2)^{-1}. \tag{32}$$

# 7 Extension, saturation, and the remaining gap

## 7.1 Removing the auxiliary restrictions

The function $q_*(n, k, \varepsilon)$ is nonincreasing in $\varepsilon$: an algorithm meeting a stricter accuracy requirement also meets every weaker one.

If $\varepsilon \geq \varepsilon_0$, then

$$c_* \min\{n, k/\varepsilon^2\} \leq (c_*/\varepsilon_0^2)k = k/1600,$$

which is below the lower bound in Proposition 2.3. Suppose next that $\varepsilon < \varepsilon_0$. If $n \geq k/\varepsilon^2$, apply Proposition 6.1. Otherwise put $\eta = \sqrt{k/n}$, so $\varepsilon < \eta$. When $\eta \leq \varepsilon_0$, monotonicity and the core lower bound at accuracy $\eta$ give

$$q_*(n, k, \varepsilon) \geq q_*(n, k, \eta) \geq c_* k/\eta^2 = c_* n.$$

When $\eta > \varepsilon_0$, instead use $n < k/\varepsilon_0^2$ and Proposition 2.3 again. This proves the second assertion of (3) for every admissible parameter triple.

To combine the two lower bounds, let $a = k(L+1)$ and $b = k/\varepsilon^2$. Since $L + 1 \leq 2^L$ for $L \geq 2$, one has $a \leq n$. If $b \geq n$, the accuracy bound already gives $c_* n$. If $b < n$, then

$$\max\{a, c_* b\} \geq \frac{c_*}{2}(a + b) \geq \frac{c_*}{2} \min\{n, a + b\}.$$

This proves Theorem 1.1. Combining its accuracy bound with Proposition 2.2 proves Corollary 1.2.

### 7.2 What the result does and does not determine

The lower bound proved in this note and the cited upper bound together give

$$c\min\{n, k(L+1) + k/\varepsilon^2\} \;\le\; q_*(n,k,\varepsilon) \;\le\; C\min\{n, kL^4/\varepsilon^3\}. \tag{33}$$

**Corollary 7.1** (An accuracy-exponent obstruction)**.** *For fixed $a \ge 0$ and $0 \le p < 2$, there is no uniform upper bound $q_* = O(\min\{n, kL^a/\varepsilon^p\})$ over the RE-03 parameter domain.*

*Proof.* Take $k = 1$, $n = 2^L$, and $\varepsilon = 2^{-L/4}$ as $L \to \infty$. Theorem 1.1 gives $q_* \ge c_* 2^{L/2}$, whereas the proposed upper expression has order $L^a 2^{pL/4}$. The ratio $2^{(2-p)L/4}/L^a$ diverges, and neither relevant expression reaches the cap $n$. This is a contradiction. □

The lower and upper bounds in (33) do not generally match. For example, for fixed positive $\varepsilon$ and increasing depth, the lower expression has order $kL$, whereas the uncapped upper expression has order $kL^4$. In a range where $\varepsilon^{-2}$ dominates $L$ and neither expression saturates at $n$, there can also be a gap in the accuracy dependence.

The construction encodes hard rank choices in a single size scale: sibling blocks of order $2k$. Their collection has packing entropy proportional to $nk$, and it yields the accuracy term $k/\varepsilon^2$. The $kL$ term comes from a separate exact-recovery family. This combination does not prove a product such as $kL/\varepsilon^2$, nor does it rule out additional joint depth–accuracy terms in the optimal complexity.

A complete solution of RE-03 would require a matching upper bound for the entire parameter range, or a stronger lower bound together with a matching algorithm. Neither is supplied here. In particular, the $n$-query baseline is only sufficient for the saturation corollary; presenting it as a full solution would leave precisely the central sublinear-query question unanswered.

## 8 Verification and reproducibility

The accompanying code is a numerical reference, not an implementation of a new sublinear-query method. It has the following components.

| File | Purpose |
|---|---|
| `code/hodlr.py` | Binary-block enumeration, blockwise SVD projection, query-counted $n$-query baseline, and hard-instance construction. |
| `code/verify.py` | Reproducible checks of algebraic identities, partition counts, objective values, and proof constants. |
| `results/verification.json` | Machine-readable output, random seed, software versions, case counts, and numerical discrepancies. |
| `results/verification.txt` | Human-readable test summary. |

Run from the package root:

```
python -m pip install -r requirements.txt
python code/verify.py --output results/verification.json
```

The test suite uses the seed 6032026. The recorded run passed 1500 local loss inequalities, 100 graph-projector identities, 12 fixed-query Gaussian KL identities, 9 partition/baseline cases, and 4 finite hard-instance checks. Rational arithmetic also checks the constants in the decoder radius, probability intersection, and saturation extension. A further Monte Carlo check uses a specified Lanczos-type adaptive query policy in four parameter configurations. Across 6000 sampled matrices and 24000 fresh query exposures, it checks the normalized Gaussian innovations and the zero-mean likelihood-ratio martingale increments. This is a check of one policy, not a verification over arbitrary adaptive algorithms.

The fixed-query KL checks explicitly form the covariance of the vectorized observations and compare its Gaussian divergence with the formula in Appendix A. They do *not* numerically test all adaptive policies. The adaptive argument, the exponentially large finite packing, and the Fano reduction are mathematical arguments in the manuscript, not conclusions inferred from experiments.

The idealized query model has exact real arithmetic and exact SVDs. NumPy uses floating-point arithmetic. Accordingly, the code's rank tests and objective comparisons use stated numerical tolerances; the exactness claims for the baseline refer to Proposition 2.2, not to bit-exact floating-point output.

## A An exact fixed-query divergence formula

For a fixed matrix $V \in \mathbb{R}^{n\times q}$ with orthonormal columns, put $R = I - VV^\mathsf{T}$. Observing $GV$ reveals exactly the component of $G$ in the Frobenius-orthogonal complement of

$$\mathcal{K} = \{RXR : X = X^\mathsf{T}\}.$$

The orthogonal projection of a symmetric shift $S$ onto $\mathcal{K}$ is $RSR$. Since the GOE has density proportional to $\exp(-n\,\|G\|_\mathsf{F}^2/4)$, its covariance is $2/n$ times the identity in an orthonormal coordinate system for symmetric matrices. Therefore

$$D_{\mathrm{KL}}(\mathcal{L}((G+\delta S)V)\|\mathcal{L}(GV)) = \frac{n\delta^2}{4}(\|S\|_\mathsf{F}^2 - \|RSR\|_\mathsf{F}^2). \tag{34}$$

This identity remains valid even though the vectorized response covariance is singular: the two laws live on the same response subspace, on which the observation map is injective for the exposed symmetric component.

In a basis adapted to $V$, write

$$S = \begin{bmatrix} S_{11} & S_{12} \\ S_{12}^\mathsf{T} & S_{22} \end{bmatrix}.$$

The norm difference in (34) is $\|S_{11}\|_\mathsf{F}^2 + 2\,\|S_{12}\|_\mathsf{F}^2$, which is at most $2\,\|SV\|_\mathsf{F}^2 \le 2q\,\|S\|_{\mathrm{op}}^2$. This recovers the bound $n\delta^2 q/2$ when $\|S\|_{\mathrm{op}} \le 1$. The calculation is a useful independent check of the normalization and the factor of two in Lemma 3.1; that lemma's adaptive proof is still necessary.

## B Checks on the logical scope of the proof

**Oracle noise versus a distribution over inputs.** The GOE matrix is sampled once and is part of the fixed input $A$. The oracle answers exact products with that same matrix on every call. No fresh noise is added to individual queries. The proof averages the fixed-input guarantee over this distribution, which is legitimate precisely because the guarantee holds for every input.

**Rank-constrained blocks versus leaves.** The hard blocks have size $2k \times 2k$ and are off-diagonal children inside nodes of size $4k$. Placing the same construction in unrestricted leaves would not force subspace selection and would not prove the accuracy lower bound.

**Global relative error versus local excess.** The decoder does not assume that each block has a relative-error guarantee. It uses the sum of nonnegative local excesses, bounded by the global excess. The squared excess factor is $(1+\varepsilon)^2 - 1 = 2\varepsilon + \varepsilon^2$, not $\varepsilon^2$.

**Symmetric inputs versus symmetric outputs.** Only the hard input distribution is symmetric. The local lemma uses the row space of an arbitrary rank-$k$ output block, so it covers nonsymmetric outputs and ranks below $k$.

**Unbounded computation and measurable decoding.** The packing is finite, and its nearest-neighbor decoder may be exponentially expensive. This is permitted because RE-03 charges only queries. Gaussian exposure and conditional non-atomicity address the stated measurable, exact-real model rather than a finite-bit surrogate.

**Sublinear upper bounds.** Neither the numerical tests nor the lower-bound construction provides an algorithm attaining (2) for all parameters. The lack of a matching upper bound is substantive, not merely an omitted implementation detail.

## References


[1] OpenProblemsInNLA repository. *RE-03: Optimal matvec query complexity of HODLR approximation.* Problem statement and audit dated September 10, 2026. `https://github.com/ajt60gaibb/OpenProblemsInNLA/tree/main/randomized-and-low-rank-approximation/RE-03`. Raw statement: `https://raw.githubusercontent.com/ajt60gaibb/OpenProblemsInNLA/main/randomized-and-low-rank-approximation/RE-03/README.md`.

[2] Tyler Chen, Feyza Duman Keles, Diana Halikias, Cameron Musco, Christopher Musco, and David Persson. *Near-optimal hierarchical matrix approximation from matrix-vector products.* Proceedings of SODA 2025, pp. 2656–2692. ArXiv:2407.04686v2, October 24, 2024. Definitions and problem: Section 1.1; upper and lower bounds: Theorems 1.1–1.2; lower-bound discussion: Section 6; outlook: Section 8. `https://arxiv.org/html/2407.04686v2`.